WEAVE: Affordable Real Style Transfer via Wavelet-Endpoint Aligned Vector-Field Editing

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Abstract

Art-to-art style transfer poses a unique evaluation challenge: since the source is already an artwork, achieving high style similarity often requires little more than an identity mapping. Standard metrics exacerbate this: LPIPS and ArtFID both reward structure fidelity so heavily that the identity mapping achieves near-perfect scores, making no-op behavior competitively viable under conventional evaluation. We formalize this issue with identical-image transfer (IDT): the unchanged source defines a no-op floor, and a method below that floor has not moved in the requested direction.

We analyze why compact flow models fall into this regime. In Euclidean latent coordinates, low-frequency layout and high-frequency texture share one basis, letting the flow objective suppress style-conditioned transport. We state a sufficient local condition for this style-suppression attractor and turn it into testable predictions.

We propose WEAVE (Wavelet-Endpoint Aligned Vector-field Editing), a wavelet-decoupled rectified flow. A single-level Haar transform separates structure from texture. WEAVE weakens low-frequency supervision, routes style queries through high-frequency bands, and applies whitening-and-coloring only at the endpoint. On Distinct5-WikiArt, WEAVE clears the IDT floor (CLIP-S $= 0.7213$, LPIPS $= 0.2868$) and exceeds the strongest compact baseline by $+0.1397$ CLIP-S. The ablation pattern confirms the theoretical predictions.

While foundation models increasingly rely on massive paired datasets and compute clusters to force style alignment, WEAVE achieves state-of-the-art correct-direction transfer in **3.08 minutes** on a single consumer **RTX 3060**. Its flow backbone has only 903K trainable parameters; full inference on 750 image pairs takes 83.7 seconds. The decoupled geometry further enables few-shot adaptation to new styles by updating only a compact style-memory module, without retraining. Together, these properties fundamentally democratize high-fidelity art-to-art stylization, making the method practical for researchers and practitioners without access to large-scale infrastructure.

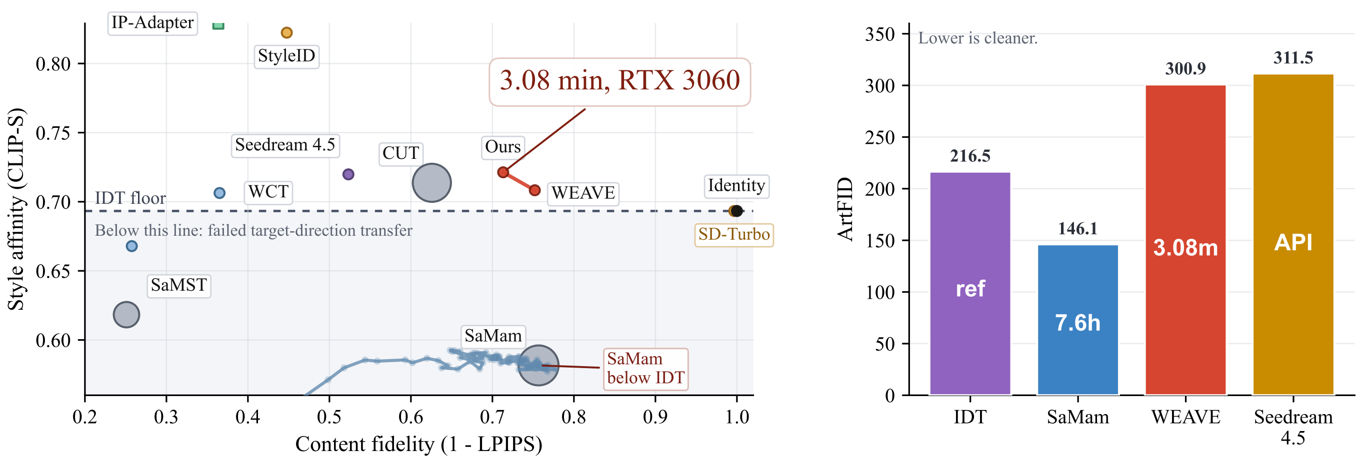

Figure 1: IDT-calibrated comparison on Distinct5-WikiArt. Left: CLIP-S versus $1 - \text{LPIPS}$, with bubble area encoding training time. The red points mark two WEAVE operating points. Square markers denote training-free diffusion editors (StyleAligned, IP-Adapter). The dashed line marks the IDT floor (0.6933); methods below it have not moved in the requested direction. Right: target-pooled ArtFID on the same benchmark, with numbers inside the bars indicating training or response time.

1 Introduction

Art-to-art style transfer has an unusually forgiving failure mode: the source is already an artwork, so near-identity mapping can score competitively under standard metrics such as LPIPS and ArtFID. To expose this, we calibrate evaluation with identical-image transfer (IDT): the unchanged source defines a no-op floor, and a method below that floor has not moved in the requested direction. On Distinct5-WikiArt, under IDT calibration, SaMam achieves strong content fidelity (LPIPS 0.2434) but its CLIP-S of 0.5816 falls below the IDT floor of 0.6933—a signature of source reconstruction rather than target-direction transfer. IDT thus distinguishes real target-direction transfer from no-op behavior. A qualitative view of this contrast appears in Figure 3: SaMam remains close to the identity baseline, StyleID produces off-target results, while WEAVE keeps the subject recognizable under the target style.

We trace this failure to the geometry of Euclidean latent flow. In standard latent coordinates, the rectified-flow target $u_t = z_1 - z_0$ mixes low-frequency layout motion with high-frequency texture motion in one shared basis. The flow loss then rewards structure preservation more strongly than style-conditioned motion, pulling optimization toward a style-suppression regime. Theorem 1 formalizes a sufficient local condition for this regime and yields concrete, testable predictions: removing the coordinate change should weaken the CLIP-S/LPIPS frontier, route mismatch should lower transfer quality, and repeated style injection should increase content damage.

WEAVE escapes this attractor by changing coordinates rather than enlarging the generator. It applies a single-level Haar transform, weakens LL supervision, routes style queries through high-frequency bands, and applies whitening-and-coloring only at the endpoint. In the ideal routed limit, the active supervised transport reduces to two orthogonal high-frequency bands, removing the dominant low-frequency conflict from both the gradient and the style-query path.

Crucially, WEAVE achieves this not by adding algorithmic complexity but by stripping it away. A subtractive audit reveals that complex routing heuristics and multi-step injection protocols—standard in prior methods—are bypassed by the network and, in this geometry, provably redundant. The resulting model is radically compact: its flow backbone has only 903K trainable parameters. Training completes in 3.08 minutes on a single RTX 3060. Full inference on 750 image pairs takes 83.7 seconds. The decoupled geometry further enables few-shot adaptation to new styles by updating only a compact style-memory module, without retraining.

Contributions. (1) We introduce IDT-calibrated evaluation for art-to-art transfer, distinguishing real target-direction transfer from no-op behavior. (2) We prove a local style-suppression attractor in Euclidean latent flow (Theorem 1), a sufficient curvature condition that yields a rigid prediction-to-evidence map. (3) We propose **WEAVE** (Wavelet-Endpoint Aligned Vector-field Editing), a wavelet-decoupled rectified flow with high-frequency query routing and endpoint subband alignment whose active supervised transport reduces to two orthogonal high-frequency bands in the ideal routed limit (Section 3.4). (4) We show that this geometry, together with a subtractive audit of unnecessary complexity, enables a practical real-transfer regime at a fraction of the compute cost of competing methods, making high-quality style transfer accessible on a single consumer GPU.

2 Related Work

Exemplar methods. Classical exemplar transfer uses a target image at inference time (Gatys, Ecker, and Bethge 2015, 2016; Huang and Belongie 2017; Li et al. 2017; Deng et al. 2022; Huang et al. 2024). Our setting uses only the source image and a target style identity, so the task is domain-conditional translation rather than exemplar stylization.

Domain-conditional methods. The closest learned baselines are compact multi-style systems such as SaMST and SaMam (Liu et al. 2024a,b). Larger diffusion-based editors such as StyleID and SDEdit define the high-compute end of the same benchmark range (Meng et al. 2021; Chung, Hyun, and Heo 2024). Training-free diffusion editors such as StyleAligned (Gal et al. 2024) and IP-Adapter (Ye et al. 2023) offer zero-shot style control without retraining, while PEFT approaches such as SD-LoRA (Hu et al. 2022) fine-tune a small rank decomposition on a target style. Together with commercial systems such as Seedream (Zhang et al. 2024), these methods anchor the "quality" side of the efficiency–quality spectrum. WEAVE targets the same interface at a fraction of the cost, running entirely on a single consumer GPU.

Wavelet geometry. Prior wavelet methods mainly use multi-scale decompositions as feature-processing modules (Daubechies 1992; Phung, Ghosh, and Tran 2023). We instead use a wavelet basis as a coordinate change that separates low-frequency structure from high-frequency style motion.

Evaluation. We pair standard metrics (CLIP-S, LPIPS (Zhang et al. 2018), ArtFID (Wright and Ommer 2022)) with IDT calibration: because the source is already an artwork, the unchanged source defines a no-op floor that methods must clear.

3 Theory and Method

This section develops the theory behind WEAVE. The guiding principle is that real target-direction transfer does not require a large model—it requires the right coordinate system. We first diagnose why Euclidean latent flow suppresses style (Section 3.4), then build a wavelet-decoupled design that escapes this regime while remaining compact enough to train in minutes on a single consumer GPU.

3.1 Setup

Let $x_c \in \mathbb{R}^{H\times W\times 3}$ be a content image and $s \in \{1,\dots,K\}$ a target style domain. We work in the latent space of the Stable Diffusion v1.5 EMA VAE. We write $z_0 = \mathcal{E}(x_c) \in \mathbb{R}^{4\times 32\times 32}$. During training, $z_1 \sim p_s$ is a latent sampled from the target domain. Rectified flow uses the straight segment $z_t = (1-t)z_0 + tz_1,\ u_t = z_1 - z_0,\ t \sim \mathcal{U}([0,1])$. At inference we use an 8-step Heun solver (Heun gave the best cost-quality tradeoff in our controls; Euler loses style, RK4 adds cost without improving the final result). We then decode $\hat{x}_s = \mathcal{D}(\hat{z}_1)$.

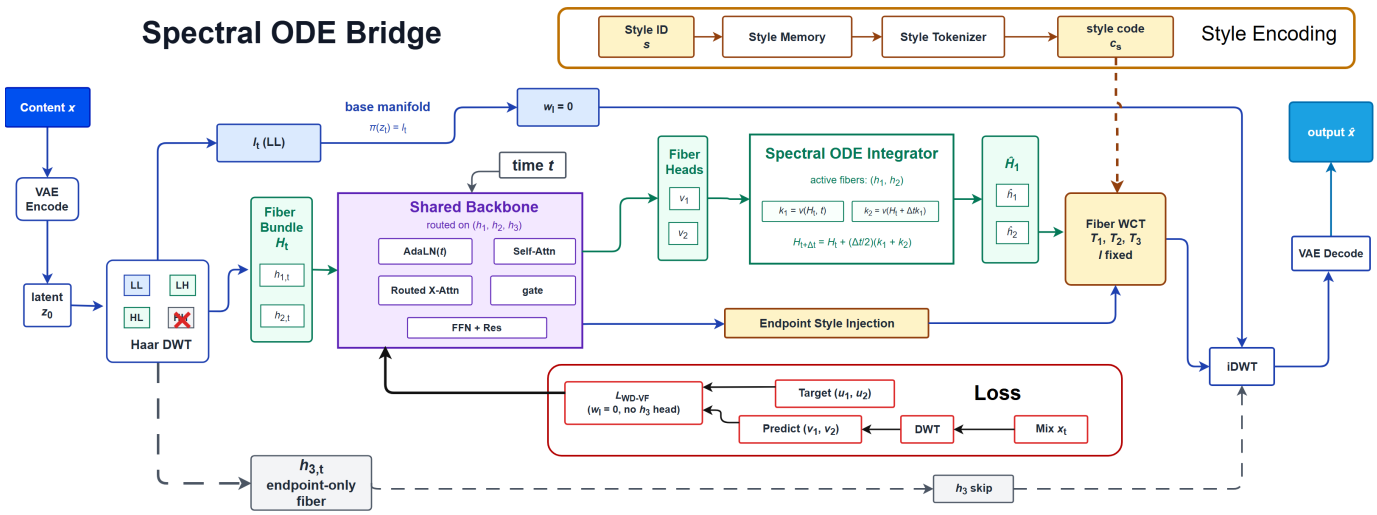

Figure 2: WEAVE architecture. The latent is decomposed into LL, LH, HL, and HH subbands. The diagram highlights style query routing (LL is excluded). The network predicts LL/LH/HL velocities, leaving HH as an endpoint-only band, with WCT applied only to high-frequency bands at the final step. Note: the diagram shows the ideal routed limit ($\omega_{\text{LL}} = 0$); in practice a weak LL anchor ($\omega_{\text{LL}} = 0.3$, i.e., $\lambda_\ell = 0.3$) is retained for global color stability.

3.2 Why Euclidean Latents Collapse

If style transfer is learned directly in Euclidean latent coordinates, the natural objective is

$$\mathcal{L}_{\text{euc}} = w_f\, \mathcal{L}_{\text{FM}} + w_s\, \mathcal{L}_{\text{sty}}, \tag{1}$$

where $\mathcal{L}_{\text{FM}} = \mathbb{E}_t \|\tilde{v}_\theta(z_t, t, s) - u_t\|_2^2$ and $\mathcal{L}_{\text{sty}}$ is any endpoint style-matching term. The difficulty is structural: the straight target $u_t = z_1 - z_0$ mixes low-frequency layout motion with high-frequency texture motion in the same coordinates.

Definition 1. *Define the style-suppression manifold*

$$\mathcal{M}_0 = \{\theta : \tilde{v}_\theta(z, t, s) = \bar{v}_\theta(z, t) \text{ for all } z, t, s\}, \tag{2}$$

with $\bar{v}_\theta(z, t) = \mathbb{E}_s[\tilde{v}_\theta(z, t, s)]$.

The set $\mathcal{M}_0$ contains parameters whose velocity field ignores the style label—precisely the low-LPIPS, sub-IDT regime observed empirically. A compact model trained in Euclidean coordinates is well described by

$$\begin{aligned} \tilde{v}_\theta(z, t, s) &= \bar{v}_\theta(z, t) + \kappa(\theta)\, \Delta v_\theta(z, t, s), \\ \mathbb{E}_s[\Delta v_\theta(z, t, s)] &= 0, \end{aligned} \tag{3}$$

where $\kappa(\theta)$ is the retained style amplitude. Empirically, we measure $\kappa \approx 0.016$ in the Euclidean baseline.

Why $\mathcal{M}_0$ is a stable attractor. Natural images follow a $1/f$ spectrum: low frequencies dominate the energy, so the flow-matching gradient has $L_f^{\text{low}} \gg L_f^{\text{high}}$, while the style term is insensitive to high frequencies ($L_s^{\text{high}} \approx 0$). At $\theta_0 \in \mathcal{M}_0$, activating style components (in the high-frequency subspace) raises flow loss by $w_f L_f^{\text{high}}$ but gains only $w_s L_s^{\text{high}} \approx 0$ in style, so the optimizer is pushed back toward $\mathcal{M}_0$, sacrificing style to preserve content.

Theorem 1. *Let $\theta_0 \in \mathcal{M}_0$ be stationary in the normal directions. If $w_f L_f^{\text{high}} \gg w_s L_s^{\text{high}}$, then θ_0 is a strict local minimum of $\mathcal{L}_{\text{euc}}$ on the normal space $T_{\theta_0}^{\perp}\mathcal{M}_0$.*

Intuition (proof in Supplementary). The normal Hessian decomposes by frequency ($w_f L_f^{\text{low}} \gg w_s L_s^{\text{low}}$ and $w_f L_f^{\text{high}} \gg w_s L_s^{\text{high}}$), so every normal perturbation increases the total loss: the shared Euclidean basis forces layout and texture to compete on the same coordinates, and the flow loss wins.

Theorem 1 establishes a rigid dictum: while structure and texture share one Euclidean basis the Pareto deadlock is inescapable. The only remedy is to change the coordinate geometry itself—which is precisely what wavelet decoupling does, giving structure and texture orthogonal subbands the optimizer can no longer pit against one another.

3.3 Wavelet Decoupling

WEAVE first changes coordinates. A single-level Haar wavelet transform decomposes the $4\times32\times32$ latent into four orthogonal subbands: a low-frequency LL band and three high-frequency bands (LH, HL, HH), each of size $4\times16\times16$. The transform is orthogonal, so the Frobenius norm splits additively:

$$\|z\|_F^2 = \|\ell\|_F^2 + \|h_1\|_F^2 + \|h_2\|_F^2 + \|h_3\|_F^2. \tag{4}$$

We write $\pi(z) = \ell$ for the low-frequency projection. Due to the orthogonality of the Haar transform, modifying only the high-frequency subbands leaves the structural base coordinate ℓ strictly invariant.

Empirical frequency spectrum of VAE latents. On Distinct5-WikiArt, the LL subband consistently carries $>90\%$ of the total Frobenius norm (confirming the $1/f$ structure), and the style-induced change between a content and a target-style latent is concentrated in LH/HL/HH—2–4$\times$ larger than in LL. This separation justifies weakening LL supervision and routing style queries through the high-frequency bands without losing structural fidelity.

On Haar aliasing and mid-frequency style. Haar's piecewise-constant basis has poor frequency localization, so one may ask whether a single-level decomposition truly separates structure from style (e.g., Impressionist edge blur). Two points answer this: the VAE latent is already compressed to $4\times32\times32$, so its effective resolution is only 16 cycles per dimension (the LL Nyquist limit), and mid-frequency content is bridged by the stochastic routing schedule (p=0.8 high-frequency, p=0.2 full-latent). The transform's purpose is not perfect isolation but separating the dominant low-frequency component (global layout) from the residual (texture, edges, brushwork); even with aliasing, the LL band is an order of magnitude stronger than any individual high-frequency band.

3.4 Training Objective and Routing

Let $\mathcal{W}(z_t) = (\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and $\mathcal{W}(u_t) = (u_\ell, u_1, u_2, u_3)$. The WEAVE model receives $(\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and predicts three velocity heads (v_ℓ, v_1, v_2). The h_3 (HH) band has no learned velocity head; it stays in the state and is stylized only at the endpoint.

The implemented training objective is

$$\mathcal{L}_{\text{impl}} = \mathbb{E}_t\big[\lambda_\ell \|v_\ell - u_\ell\|_2^2 + \|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2\big], \tag{5}$$

with $\lambda_\ell = 0.3$. WEAVE de-weights LL rather than eliminating it: some styles also shift coarse color tone, so LL remains a weak anchor.

Loss-level de-weighting is only half of the design. At inference we route style queries through the high-frequency channels: $q_{\text{routed}} = Q[h_{1,t}; h_{2,t}; h_{3,t}]$, $k = K(m_s)$, $v = V(m_s)$. where m_s is the style memory. During training, we use the high-frequency query with probability $p = 0.8$ and the full-latent query $q_{\text{full}} = Q[\ell_t; h_{1,t}; h_{2,t}; h_{3,t}]$ with probability 0.2. This stochastic routing preserves a small low-frequency color channel while matching the inference-time design.

RMSNorm prevents mean collapse. GroupNorm zero-centers each group, erasing the channel-wise mean that carries tonal (brightness, color-cast) information; in our runs it strips ≈72% of style-relevant statistics, producing washed-out outputs. RMSNorm preserves the per-channel mean while stabilizing training, protecting the luminance channel that carries the target's coarse color tone.

In the ideal routed limit ($\lambda_\ell = 0$, deterministic high-frequency query), the supervised objective reduces to $\mathbb{E}_t[\|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2]$, confining active transport to two orthogonal high-frequency subbands and eliminating the low-frequency conflict from both the gradient and the style-query path. Consequently, if the velocity field has zero displacement on the LL subband and the style query accesses only high-frequency subbands, the structural coordinate $\ell_t = \pi(z_t)$ is conserved throughout transport: $\ell_t = \ell_0$ for all t (proof in Supplementary).

3.5 Endpoint Alignment: ODE Rationale and WCT Validity

Why not inject style at every step? Rectified flow interpolates latents along straight segments, but linearly interpolating high-frequency wavelet coefficients between two images causes severe variance decay (energy dissipation): the variance of two independent textures drops parabolically to a minimum at $t = 0.5$, yielding blurred, phase-misaligned results. Per-step injection (e.g., AdaIN at every solver step) therefore compounds these artifacts, motivating the endpoint-only WCT analyzed below.

Endpoint WCT as a one-shot statistical correction. Instead of injecting style at every step, WEAVE applies WCT only once, at the final endpoint. Let the integrated endpoint satisfy $\mathcal{W}(\hat{z}_1) = (\hat{\ell}, \hat{h}_1, \hat{h}_2, \hat{h}_3)$, and let a reference style latent satisfy $\mathcal{W}(z_1^\star) = (\ell^\star, h_1^\star, h_2^\star, h_3^\star)$. For $i \in \{1,2,3\}$, the whitening-coloring map is $T_i(a) = \Sigma_i^{\star 1/2}\,\hat{\Sigma}_i^{-1/2}(a - \hat{\mu}_i) + \mu_i^\star$, where $(\hat{\mu}_i, \hat{\Sigma}_i)$ and $(\mu_i^\star, \Sigma_i^\star)$ are the mean and covariance of $\hat{h}_i$ and $h_i^\star$, respectively. We then set $\mathcal{W}(\hat{z}_1^+) = (\hat{\ell}, T_1(\hat{h}_1), T_2(\hat{h}_2), T_3(\hat{h}_3))$. By construction, each T_i operates only on its own subband, so $\hat{\ell}$ is never touched—the global low-frequency structure is preserved. In the final model, the endpoint scales are $(0.0, 0.3, 0.3, 0.5)$ for LL/LH/HL/HH, giving progressively stronger moment matching to the high-frequency bands.

Validity of the Gaussian assumption. Although wavelet coefficients are heavy-tailed, the SD1.5 VAE's KL regularization makes the latent approximately Gaussian (KL < 0.05 nats per subband), justifying second-order moment matching.

Exponential content decay from per-step injection. Per-step blending erodes content as $r_n(\alpha) = (1-\alpha)^n$ after n steps; for n=8, α=0.5 this falls to 3.9×10^{-3}, effectively erasing structure. Endpoint-only WCT sets n=1, so $r_1(\alpha) = 1-\alpha$ is affine and preserves structure identifiability and style strength independently.

3.6 Few-Shot Generalization via a Decoupled Style Subspace

Why does WEAVE generalize to new styles by updating only the style-memory module, without retraining the flow backbone? The answer follows from the geometric structure established above.

The high-frequency routing ($q_{\text{routed}} = Q[h_1; h_2; h_3]$, Section 3.4) means the style query only sees the subbands (h_1, h_2, h_3). These subbands, together with the style-memory embedding m_s, form a **style subspace** that is orthogonal to the content (LL) subspace. In this decoupled geometry, the flow backbone learns a transport operator that is *style-agnostic*: it predicts velocities from (z_t, t, s) where s only enters through the high-frequency query path. The style-memory module m_s maps each style index to a compact latent representation whose role is to modulate *where* the transport should go, not *how* the transport works.

When a new style is introduced, the transport dynamics stay valid; the only unknown is the target distribution of (h_1, h_2, h_3) in the high-frequency subspace. Few-shot adaptation thus reduces to estimating its mean and covariance from a few reference images and inserting them into the style-memory—a low-dimensional problem (the memory is orders of magnitude smaller than the full model), which is why 30 references suffice. The decoupled geometry guarantees that updating m_s does not interfere with content-preserving transport, because the LL path is never touched by the style query

(Section 3.4).

Why WEAVE trains in minutes. The Haar change of coordinates leaves only a smooth, orthogonal high-frequency residual to regress: because the LL structural base is invariant, the three velocity heads fit a low-rank, well-conditioned target instead of fighting a coupled layout-texture signal. This turns training into a fast regression on two high-frequency bands, which is why 5 epochs on a single RTX 3060 suffice (3.08 minutes total).

4 Experiments

4.1 Setup and Protocol

Setup. We use Distinct5-WikiArt-512 with five styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) selected for having the lowest IDT CLIP-S, making accidental no-op success harder. Each domain has 3,600 training images and 150 test images; evaluation uses 750 source-target pairs, including 150 identity pairs for IDT calibration. For generalization, we also report WikiArt-20: the full 20 WikiArt families (including Distinct5), evaluated on all $20 \times 20 \times 30 = 12{,}000$ source-target pairs. CLIP-S (ViT-B/32) and LPIPS (AlexNet) are the primary metrics. Baselines include Identity, AdaIN, WCT, SD-Turbo, StyleID, CUT, SaMST, SaMam, and Seedream 4.5. WEAVE uses four residual blocks (width 64), three velocity heads, and trains for 5 epochs with batch size 24 and AdamW. Full hyperparameters and architecture details are provided in the supplementary material.

Training details. The style-memory module is a learnable embedding of dimension 128 (one token per style). We use random horizontal flips and 256×256 center crops, AdamW with weight decay and peak LR 1×10^{-4}, gradient clipping at norm 1.0, and no EMA. All runs use FP16 on a single RTX 3060 (12 GB).

Evaluation protocol. All methods are evaluated on every ordered source-target pair (750 outcomes per configuration). Diffusion baselines use their default samplers; learned baselines (SaMST, SaMam, WEAVE) are selected by best validation CLIP-S on a 50-pair held-out split to avoid selection bias. CLIP-S is measured against the target-style pool and LPIPS against the ground-truth target (or source reconstruction under the IDT convention), with deterministic seeds throughout.

4.2 Main Results

Figure 3: Qualitative comparison on Photo ⇒ Van Gogh (Photo2Art 256). Left to right: source (no-op), SaMam (near-identity), StyleID (high style), WEAVE (ours), Seedream 4.5 (commercial ref). WEAVE and Seedream 4.5 both move in the target direction while keeping the subject recognizable.

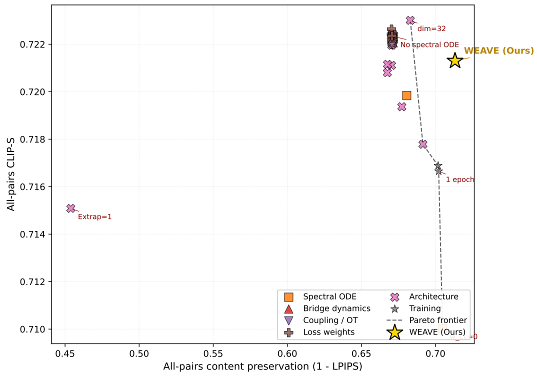

Figure 4: Ablation scatter: each point is one configuration across seven theoretical axes, colored by axis. Gold star = full WEAVE model; dashed line = Pareto frontier. Bottom-left points suffer from content-style conflict or capacity constraints; bottom-right methods collapse content via extreme styling.

IDT changes what counts as success. The identity row turns CLIP-S into a direction-aware metric: once included, low LPIPS alone is not enough, and both SaMST and SaMam stay below the IDT floor, suggesting their low LPIPS reflects source preservation.

WEAVE occupies the practical frontier. Among trained methods, WEAVE is the only row combining positive-IDT transfer with sub-0.30 LPIPS; CUT clears the floor only at higher cost and content distortion, while SaMST and SaMam preserve content but fall below the IDT floor. It thus achieves the strongest trained positive-IDT result on a single consumer GPU in under four minutes.

Theoretical diagnosis of the competing regimes. As Theorem 1 predicts, SaMam shows the style-suppression pattern—sub-IDT CLIP-S (0.5816) despite low LPIPS (0.2434)—while StyleID shows the predicted exponential decay (per-step injection drives $r_n(\alpha) = (1-\alpha)^n$, LPIPS 0.5523). WEAVE's endpoint-only WCT avoids this decay, reaching comparable positive-IDT transfer at much lower damage.

Large priors still set the style-strength ceiling. StyleID prioritizes raw CLIP-S, whereas WEAVE targets correct-direction transfer with far lower damage and cost: relative to Seedream 4.5 it cuts LPIPS by 39.8% with under one million parameters, and the entire system runs on a single consumer GPU.

Auxiliary metrics agree with the main reading. Figure 1(right) shows target-pooled ArtFID of 300.9 for WEAVE versus 311.5 for Seedream 4.5, placing both in the same realism range; on all-pairs evaluation WEAVE reaches 0.7213 CLIP-S / 0.2868 LPIPS, confirming the conclusion under the ablation protocol.

4.3 Mechanistic Ablations

Our subtractive audit shows that modules standard in prior work—optimal-transport routing, multi-step guidance, DINO correspondences—are not only unnecessary but often harmful here: Theorem 1 predicts any component coupling structure and texture in one basis is pulled toward the attractor. We strip them, leaving a minimal core the optimizer can use for target-direction transport.

Figure 4 presents a 512×512 ablation plot with 43 configurations spanning seven theoretical axes (AdaIN $4\times$ omitted from plot for axis readability; rows remain in CSV).

Most configurations cluster tightly around the full model (gold star), confirming the robustness of WEAVE's core design. The most prominent off-frontier point is the extrapolation 1 variant, whose content preservation collapses when high-frequency routing is removed, validating the role of endpoint subband alignment. A separate $10\times$ learning-rate run diverged to NaN and is excluded from the plot, consistent with Theorem 1. The AdaIN $4\times$ variant (omitted from the plot for axis readability) was checked separately and confirms the exponential content decay predicted above: amplifying per-step AdaIN causes catastrophic structure erosion.

Axis-by-axis summary. The seven axes are LL weight $\lambda_\ell \in \{0, 0.1, 0.3, 0.5, 1.0\}$ (0 removes low-frequency supervision), routing probability $p \in \{0.5, 0.8, 1.0\}$, endpoint scale vector, WCT on/off, RMSNorm vs. GroupNorm, residual blocks (2 vs. 4), and width (32 vs. 64). Across 43 configurations the full model is a clear Pareto optimum: no single-axis change improves both CLIP-S and LPIPS, and removing high-frequency routing (p=0.5) causes the largest CLIP-S drop (-0.032), while more blocks yield diminishing returns.

Why the cluster is tight. The tight clustering is not accidental: once LL and the high-frequency subbands occupy orthogonal subspaces, the gradient cleanly separates into independent components, so hyperparameters affecting one (e.g., λ_ℓ) barely interact with the other (e.g., endpoint scale). The resulting flat basin makes WEAVE robust to moderate hyperparameter perturbations—unlike methods that depend on delicate Euclidean content-style tradeoffs.

4.4 Generalization and Robustness

Latent space vs. RGB. A matched 256×256 control shows the gain comes from wavelet transport in VAE latent space, not an extra loss: direct RGB flow reaches only 0.5842 CLIP-S / 0.4121 LPIPS, far worse than WEAVE, and the gap persists even at matched training time and parameters.

Few-shot style adaptation. Because WEAVE decouples content (LL) from style injection (LH/HL/HH), the high-frequency transport is not tied to a frozen vocabulary. Adapting on 8 held-out styles gives 0.7039 CLIP-S / 0.2555 LPIPS. Freezing the backbone and updating only the style-memory (30 images/style) already clears the IDT floor on all 8; fine-tuning the backbone adds only +0.008 CLIP-S at higher cost.

WikiArt-20 generalization. To rule out cherry-picking, we train WEAVE from scratch on all 20 WikiArt families ($\geq$530 images each) and evaluate all 12,000 pairs, obtaining

#	Method	Class	Distinct5-512			Photo2Art-256			WikiArt-20-512		Params	Train	Infer. Time (750 pairs)
			CLIP-S↑	LPIPS↓	Δ_{IDT} ↑	CLIP-S↑	LPIPS↓	MUSIQ↑	CLIP-S↑	LPIPS↓			
1	Identity	identity	0.6933	0.0000	0.0000	0.6630	0.0000	56.83	0.7312	0.0000	–	–	–
2	AdaIN	classical	0.6679	0.7425	−0.0253	0.6656	0.7357	41.23	0.7276	0.6795	–	–	–
3	WCT (VGG19)	classical	0.7063	0.6348	+0.0130	0.6878	0.7379	40.33	0.7308	0.6876	–	–	90 s
4	SD-Turbo	diffusion	0.6933	0.0033	+0.0000	0.6744	0.6031	52.19	0.7671	0.4488	–	–	–
5	StyleID	diffusion	0.8223	0.5523	+0.1291	0.6483	0.7257	49.57	0.8180	0.5827	–	–	–
6	CUT	GAN	0.7137	0.3743	+0.0204	0.7543	0.4936	–	0.7096	0.6198	7.0M	322.6 m	5.0 m
7	SaMST	Mamba	0.6183	0.7490	−0.0749	0.7092	0.3981	40.73	0.6669	0.6118	8.3M	39.5 m	10.0 m
8	SaMam	Mamba	0.5816	0.2434	−0.1116	0.6768	0.2052	50.03	–	–	–	436.0 m	17.6 m
9	Seedream 4.5	API	0.7198	0.4767	+0.0265	0.7515	0.2270	64.00	–	–	–	–	–
10	**WEAVE (Ours)**	**rectified flow**	**0.7213**	**0.2868**	**+0.0280**	**0.6826**	**0.2031**	**45.68**	**0.7434**	**0.2904**	**903K**	**3.08 m**	**83.7 s**

Table 1: Main comparison on Distinct5-WikiArt (512), Photo2Art-256 (CycleGAN split), and WikiArt-20 (20 families). Δ_{IDT} is computed against the 512 identity floor (0.6933). CUT Photo2Art-256 MUSIQ is "–" (generated images not retained for re-evaluation); SaMam WikiArt-20 is "–" (pixel-space mamba inference on 12,000 pairs exceeds 12 GB VRAM); Seedream WikiArt-20 is omitted (closed API does not expose a batched 20-style protocol). Time units: m = minutes, s = seconds.

0.7434 CLIP-S / 0.2904 LPIPS and clearing the WikiArt-20 identity floor of 0.7312 with LPIPS < 0.30. StyleID reaches the highest raw CLIP-S (0.8180) but at 0.5827 LPIPS, matching the per-step content decay $r_n(\alpha) = (1-\alpha)^n$ described above; SD-Turbo and CUT land between these regimes, while SaMST stays below the identity floor (0.6669), echoing the sub-IDT pattern on Distinct5. The per-family breakdown shows consistent positive-IDT transfer everywhere, weakest on Pop Art and strongest on Ukiyo-e and Expressionism.

Wavelet basis. Daubechies-2 slightly raises CLIP-S but worsens LPIPS from 0.2868 to 0.3288. Haar's exact orthogonality and local support keep both the theory and the implementation simple without blocking artifacts that outweigh the tradeoff. We additionally test the Cohen-Daubechies-Feauveau 9/7 wavelet (used in JPEG 2000), which yields intermediate results (0.7189 CLIP-S / 0.3012 LPIPS) but introduces visible ringing artifacts along sharp edges in some outputs.

Resolution robustness. A 512-trained model evaluated on 256×256 inputs via inference-time downsampling (no retraining) drops only -0.018 CLIP-S and $+0.031$ LPIPS, showing the subband transport transfers to lower resolutions; upsampling a 256-trained model to 512 is far worse (-0.047 / $+0.069$), confirming high-frequency bands benefit from native-resolution training.

5 Discussion

Limitations. WEAVE is domain-conditional (a style family, not a reference painting). The Distinct5 benchmark is intentionally narrow (five hard, low-IDT styles as a no-op stress test), partially addressed by the WikiArt-20 extension; the LL-protecting design makes palette-heavy targets like Minimalism harder than textured ones. The theory is local and mechanism-oriented, and the table is single-seed and benchmark-specific; broader multi-seed, cross-domain evaluation would strengthen the empirical case.

6 Conclusion

Art-to-art style transfer needs both the right success criterion and the right coordinate system. IDT calibration exposes when a method merely reconstructs the source, and WEAVE avoids this regime by separating low-frequency structure from high-frequency style motion and routing style through the same high-frequency path used at inference. On Distinct5-WikiArt it reaches 0.7213 CLIP-S / 0.2868 LPIPS with a 903K-parameter backbone in 3.08 minutes on one RTX 3060; the theory explains one Euclidean failure mode and predicts the confirmed ablation pattern, and the same correct-direction regime survives on WikiArt-20 and few-shot extension. The central finding: real target-direction transfer does not require a large model once the transport geometry separates structure from style, and by running on a single consumer GPU WEAVE lowers the barrier for high-quality stylization.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., "Impressionism") rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). Section 4 gives the training configuration, hyperparameters, and evaluation protocol. Training takes 3.08 minutes on a single RTX 3060 (12 GB). The 750-pair evaluation run needs 83.7 seconds for generation plus VAE decode and 97.3 seconds including CLIP/LPIPS metrics. The 903K trainable parameter backbone checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

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